

Element K: Reflection on the design project

Research:

The research done about this project included comprehensive compilation of key information about the habits of komodo dragons, durability of our materials, prior attempts, and techniques that may be helpful in the solution of the problem. Starting from general information about komodo dragons, we found information about the types of dragons in the Nashville Zoo. After realizing there were a variety of dragons with a variety of sizes and weights, we acknowledged the need for a variable design that could be manipulated to fit the needs of various situations. Then, we looked at the habits of komodo dragons and the safety considerations that we needed to consider. We identified the relative strength or force applied by the dragons as well as their ability to breath in low air-flow environments. We additionally researched prior attempts to the project. Although Brentwood High School has never attempted this problem, we found similar situations and examples online that we could use to guide our project. The Nashville Zoo used a metal box with holes to carry smaller dragons, Disney created a metal and plastic box with complicated mechanisms, and many large metal boxes have been used to transport crocodiles. Using these inspirations and techniques, we compiled a general plan and design to present to the zookeepers as a pitch.

Once we did this initial research, we organized an online interview with Nick Hanna who served as our expert for the majority of the project. From this interview, we were able to identify key aspects of the project that needed to be addressed -- one of which was medical imaging which became our key concern as other groups addressed the other concerns. Additionally, we were given suggestions on materials and possible mechanisms that would be preferred by the zookeepers.



Using information from the interview, we began to research materials and the amount of force those materials could maintain. Hanna suggested that we focus on using plastics for our main portion of the project so that the project could be medically imaged without artifacts appearing on the films. Starting from that, our group identified acrylic, plexiglass, and PLA as our best choices for materials.

Planning:

Using the design that we had compiled from prior attempts, the entire Komodo dragon team met to divide our project into three subparts that we could split up and do. Using the priorities of the zookeepers as guidance, we divided the project into a frame, panels, and medical imaging. Our subpart was tasked with troubleshooting medical imaging. This problem included troubleshooting x-ray, ultrasound, and blood-testing for the dragons.

Acknowledging our main concerns, we compiled a list of criteria and constraints for our task. By ranking them based on priority, we identified our most important priorities and possible weights of our tasks. Eventually, the following criteria and relative weights (out of 1) were solidified:

Simplicity of latching/opening - 0.15, Latching ability - 0.15, X-ray Compatibility - 0.25, Range of operator motion - 0.2 1/3, Durability - 0.15, Reliability - 0.1. These criteria were our guiding principles for the rest of the decisions we made. After this, we brainstormed multiple ideas for latching and compliant mechanisms. By creating multiple designs that we could compare and then placing them into a design matrix, we were able to narrow down specifics on how we planned to address the closing, locking, and opening mechanism.

Prototyping:

To start our prototyping, we created a cardboard model to simulate the guillotine style mechanism that was decided as a group effort. We learned from this model

that we needed a way to maintain friction between the surfaces, a

proper handle, and a way to open the mechanism with one hand. Soon after, we began construction on a plastic prototype using 3d elements as our tracks and holds for the ultrasound door. Then, we started designing the latching mechanism. Using expert opinions, we decided on a compliant mechanism with a ratcheting system created through 3d printed pieces. Once these pieces were designed, we attached them and the tracks to our plastic via nylon screws. Eventually, this process was recreated four more times on larger panels used on the box.

